

Supporting Information for “A General Model for Estimating Partition Coefficients to Organisms and their Tissues using the Biological Compositions and Polyparameter Linear Free Energy Relationships”

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SI-1 Comparison between the PP-LFER-composition-based-model and a tissue-specific PP-LFER

Tissue-specific PP-LFERs as published in refs 1-6 may work well for volatile compounds for which these PP-LFERs have been calibrated, but they can induce larger prediction errors if used to extrapolate partition coefficients beyond the calibration domain. This issue has been discussed previously using experimental data of milk-water partition coefficients.⁷ Below, some additional examples for this issue are provided.

We use brain-water partition coefficients ($K_{\text{brain/water}}$) as example, but similar results can be obtained by using other tissue-water partition coefficients as well.

Abraham et al. calibrated a PP-LFER model using in vitro measured brain-air partition coefficients ($K_{\text{brain/air}}$) and reported the following equation:

$$\log K_{\text{brain/air}} = -0.987 + 0.263E + 0.411S + 3.358A + 2.025B + 0.591L \quad (\text{S1})$$

E , S , A , B , and L are excess molar refraction, dipolarity/polarizability parameter, solute H-bond acidity, solute H-bond basicity; and the logarithm of the hexadecane-gas partition coefficient, respectively. For a comparison purpose, we convert $K_{\text{brain/air}}$ calculated with eq S1 to $K_{\text{brain/water}}$:

$$K_{\text{brain/water}} = K_{\text{brain/air}} K_{\text{air/water}} \quad (\text{S2})$$

$K_{\text{air/water}}$ was calculated using the PP-LFER for $K_{\text{air/water}}$ (see Table 1 of the main manuscript). This conversion unlikely causes large systematic errors, as explained in the main manuscript.

Using eqs S1 and S2, $K_{\text{brain/water}}$ values were calculated for diverse neutral chemicals including simple monofunctional compounds,^{8,9} polycyclic aromatic hydrocarbons,⁹ PCBs,¹⁰ pesticides,¹¹ and pharmaceuticals and hormones¹² (the cited references include PP-LFER descriptors used). These $K_{\text{brain/water}}$ values are compared with those calculated with the PP-LFER composition based model. The results are shown below.

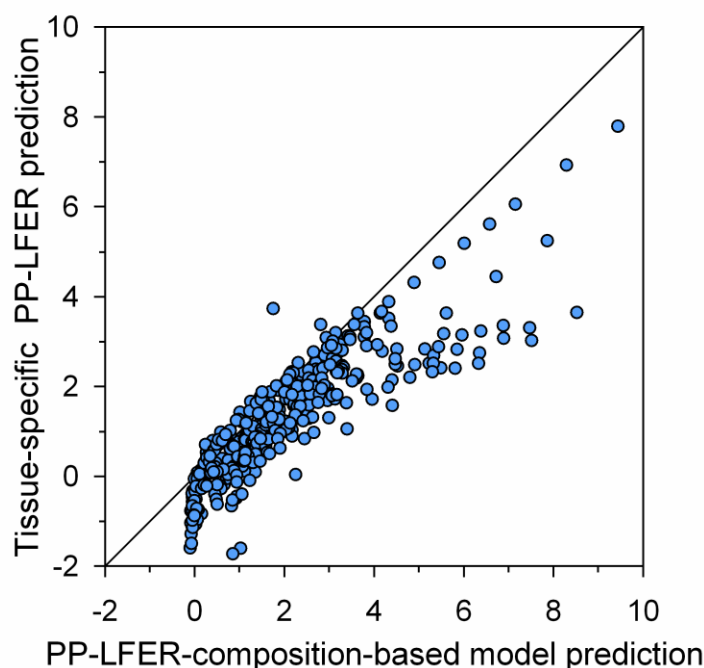


Figure SI-1. Predicted $K_{\text{brain/water}}$ using tissue-specific PP-LFER and PP-LFER-composition-based model.

Relatively good agreement was found for a number of simple monofunctional compounds (i.e., many data points in the $K_{\text{brain/water}}$ range of 0–4. This is reasonable because the brain-specific PP-LFER was calibrated with the data for such simple compounds. However, large differences were found for a number of other compounds. In particular, hydrophobic compounds ($K_{\text{brain/water}} > 4$) showed tremendously large and systematic differences (i.e., 2–5 log units) between predictions by the brain-specific PP-LFER and the PP-LFER-composition-based model. For example, $\log K_{\text{brain/water}}$ of PCB 156 is predicted as 3.5 by the brain-specific PP-LFER and 6.5 by the PP-LFER-composition-based model. For comparison, the lipid-octanol model gives 5.8 for $\log K_{\text{brain/water}}$ of PCB 156, using a brain lipid content of 11%¹³ and a $\log K_{\text{ow}}$ of 6.8 for PCB 156 (as predicted by the PP-LFER). Unfortunately, no experimental data for $\log K_{\text{brain/water}}$ of PCB 156 is available. What could be compared may be lipid-normalized bioconcentration factors (BCFs) and necroconcentration factors (NCFs) for many aquatic organisms reported by van der Heijden and Jonker.¹⁴ The reported lipid-normalized BCFs and NCFs for PCB 156 are 7.2–7.8 in log units, which are in good agreement with the lipid-normalized $K_{\text{brain/water}}$ predicted by the PP-LFER-composition-based model. Though it is possible that lipids of brain and aquatic organisms have slightly different sorption capacity for PCB 156, it is unlikely that $\log K_{\text{brain/water}}$ of PCB 156 could be as low as 3.5. Other examples of chemicals that show systematically large discrepancy between the two models are highly hydrophilic compounds. For example, $\log K_{\text{brain/water}}$ of caffeine is predicted to be –1.3 by the brain-specific PP-LFER and 0.0 by the PP-LFER-composition-based model. As explained in the manuscript, very hydrophilic compounds

simply partition into the water phase of a tissue. Brain contains 79% water,¹³ and this means that $K_{\text{brain/water}}$ cannot be lower than 0.79 (i.e., $\log K_{\text{brain/water}}$ cannot be lower than -0.1). Composition-based models can reproduce this lower boundary for hydrophilic compounds, but tissue-specific PP-LFERs cannot do so, as shown in the above figure.

SI-2 Addition of a protein term that contains f_{protein} and K_{ow} to lipid-octanol model

Some researchers proposed to add a protein term containing f_{protein} and K_{ow} to eq 7 of the manuscript (e.g., refs 15, 16).

Firstly, it should be noted that $\log K_{\text{protein/water}} - \log K_{\text{ow}}$ regression equations derived in the past are generally based on experimental data for serum albumin. Very recently, we systematically investigated $K_{\text{protein/water}}$ for several proteins and revealed that $K_{\text{protein/water}}$ for serum albumin is substantially higher than those for structural proteins.¹⁷ Previous regression models for $K_{\text{protein/water}}$ therefore are not suitable for modeling sorptive capacity of tissue proteins. Alternatively, one could use $K_{\text{protein/water}}$ for muscle proteins investigated recently by us.¹⁷ For example, a regression equation for chicken muscle protein appears,

$$K_{\text{protein/water}} = 0.73 \log K_{\text{ow}} - 0.40 \quad (\text{S3})$$

Using this in combination with the lipid-octanol model would lead to,

$$K_{\text{tissue/water}} = f_{\text{lipid}} K_{\text{ow}} + f_{\text{protein}} \cdot 10^{-0.40} \cdot K_{\text{ow}}^{0.73} + f_{\text{water}} \quad (\text{S4})$$

Use of a regression equation for fish muscle protein would provide a similar equation.

Another common approach for taking into account the sorptive capacity of proteins is to assume that $K_{\text{protein/water}}$ is 1/20 of K_{ow} .¹⁶ This would lead to the following equation,

$$\begin{aligned} K_{\text{tissue/water}} &= f_{\text{lipid}} K_{\text{ow}} + f_{\text{protein}} \cdot 0.05 K_{\text{ow}} + f_{\text{water}} \\ &= (f_{\text{lipid}} + 0.05 f_{\text{protein}}) K_{\text{ow}} + f_{\text{water}} \end{aligned} \quad (\text{S5})$$

The models described above were used to calculate $K_{\text{muscle/water}}$ and compared with the lipid-octanol (+ water) model (eq 7 in the manuscript). We chose muscle because it is the leanest of the tissues considered. As shown below in the figure, differences are marginal, up to 0.35 log units and are very systematic. Therefore, we do not show additional analysis using these protein-amended lipid-octanol models in the manuscript.

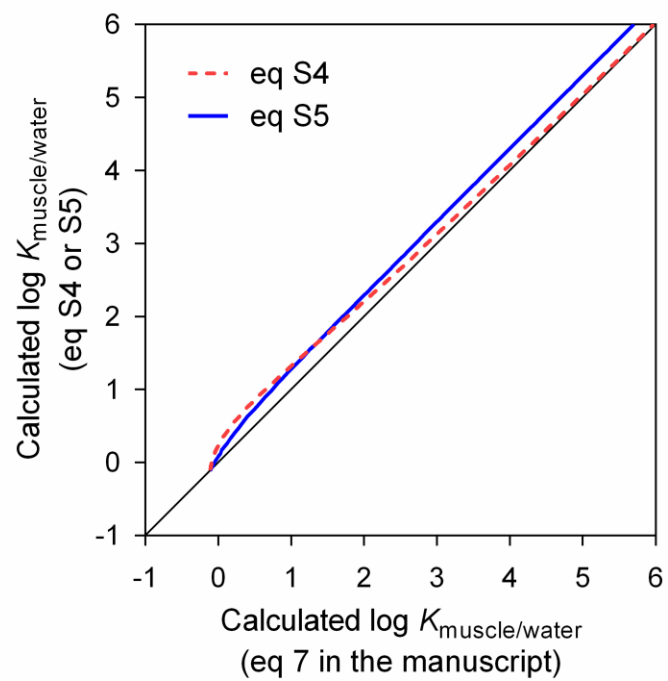


Figure SI-2. $K_{\text{muscle/water}}$ calculated with eqs 4 and 5 (protein considered) and with eq 7 (protein not considered).

Table S1. PP-LFER system coefficients that use *L* instead of *E* descriptor (L/L). Values are for 37 °C.

	<i>c</i>	<i>s</i>	<i>a</i>	<i>b</i>	<i>v</i>	<i>l</i>	SD	<i>n</i>	ref
storage lipid-water	0.55	-1.62	-1.93	-4.15	1.99	0.58	0.20	247	18
membrane-water	0.53	-0.93	-0.18	-3.75	1.73	0.49	0.28	131	19
muscle protein-water	-0.94	-0.59	0.21	-3.17	2.13	0.33	0.23	46	17
serum albumin-water	0.35	-0.46	0.20	-3.18	1.84	0.28	0.41	82	20
water-air	-0.63	2.26	3.60	4.37	-2.21	0.31	0.18	374	^a

^a Determined here using data from ref 21.

Table S2. PP-LFER descriptors used for prediction of in vitro partition coefficients.

		<i>E</i>	<i>S</i>	<i>A</i>	<i>B</i>	<i>V</i>
7440-01-9	neon	0.00	0.00	0.00	0.00	0.085
10024-97-2	nitrous oxide	0.07	0.35	0.00	0.10	0.281
100-41-4	ethylbenzene	0.61	0.51	0.00	0.15	0.998
100-42-5	styrene	0.85	0.65	0.00	0.16	0.955
10043-92-2	radon	0.00	0.00	0.00	0.00	0.384
100-80-1	m-methylstyrene	0.87	0.65	0.00	0.18	1.096
103-65-1	propylbenzene	0.60	0.50	0.00	0.15	1.139
106-42-3	p-xylene	0.61	0.52	0.00	0.16	0.998
106-93-4	1,2-dibromoethane	0.75	0.76	0.10	0.17	0.740
106-94-5	propyl bromide	0.37	0.40	0.00	0.12	0.706
106-97-8	butane	0.00	0.00	0.00	0.00	0.672
106-99-0	1,3-butadiene	0.32	0.23	0.00	0.10	0.586
107-04-0	1-bromo-2-chloroethane	0.08	0.70	0.10	0.09	0.688
107-05-1	allyl chloride	0.33	0.56	0.00	0.05	0.611
107-06-2	1,2-dichloroethane	0.42	0.64	0.10	0.11	0.635
107-83-5	2-methylpentane	0.00	0.00	0.00	0.00	0.954
107-87-9	2-pentanone	0.14	0.68	0.00	0.51	0.829
107-98-2	1-methoxy-2-propanol	0.22	0.54	0.31	0.82	0.790
108-03-2	1-nitropropane	0.24	0.95	0.00	0.31	0.706
108-10-1	4-methyl-2-pentanone	0.11	0.65	0.00	0.51	0.970
108-21-4	isopropyl acetate	0.06	0.57	0.00	0.47	0.888
108-38-3	m-xylene	0.62	0.52	0.00	0.16	0.998
108-67-8	1,3,5-trimethylbenzene	0.65	0.52	0.00	0.19	1.139
108-86-1	bromobenzene	0.88	0.73	0.00	0.09	0.891
108-87-2	methylcyclohexane	0.24	0.06	0.00	0.00	0.986
108-88-3	toluene	0.60	0.52	0.00	0.14	0.857
108-90-7	chlorobenzene	0.72	0.65	0.00	0.07	0.839
109-59-1	2-isopropoxyethanol	0.20	0.48	0.21	0.91	0.931
109-60-4	propyl acetate	0.09	0.60	0.00	0.45	0.888
109-66-0	pentane	0.00	0.00	0.00	0.00	0.813
109-69-3	1-chlorobutane	0.21	0.40	0.00	0.10	0.795

109-86-4	2-methoxyethanol	0.27	0.50	0.30	0.84	0.649
109-93-3	divinyl ether	0.26	0.39	0.00	0.13	0.645
109-94-4	ethyl formate	0.15	0.66	0.00	0.38	0.606
110-00-9	furan	0.37	0.53	0.00	0.13	0.536
110-19-0	isobutyl acetate	0.05	0.57	0.00	0.47	1.028
110-43-0	2-heptanone	0.12	0.68	0.00	0.51	1.111
110-54-3	hexane	0.00	0.00	0.00	0.00	0.954
110-80-5	2-ethoxyethanol	0.24	0.50	0.30	0.83	0.790
110-82-7	cyclohexane	0.31	0.10	0.00	0.00	0.845
111-27-3	1-hexanol	0.21	0.42	0.37	0.48	1.013
111-65-9	octane	0.00	0.00	0.00	0.00	1.236
111-66-0	1-octene	0.09	0.08	0.00	0.07	1.193
111-76-2	2-butoxyethanol	0.20	0.53	0.26	0.83	1.071
111-84-2	nonane	0.00	0.00	0.00	0.00	1.377
1120-21-4	undecane	0.00	0.00	0.00	0.00	1.659
112-40-3	dodecane	0.00	0.00	0.00	0.00	1.799
115-07-1	propene	0.10	0.08	0.00	0.07	0.488
115-10-6	dimethyl ether	0.00	0.27	0.00	0.41	0.449
119450-77-0	1,1,2,4,4-pentafluorobutane	-0.49	0.48	0.23	0.14	0.861
123-51-3	3-methyl-1-butanol	0.19	0.39	0.37	0.48	0.872
123-86-4	butyl acetate	0.07	0.60	0.00	0.45	1.028
123-91-1	1,4-dioxane	0.33	0.75	0.00	0.64	0.681
123-92-2	isopentyl acetate	0.05	0.57	0.00	0.47	1.169
124-11-8	1-nonene	0.09	0.08	0.00	0.07	1.334
124-18-5	decane	0.00	0.00	0.00	0.00	1.518
124-48-1	chlorodibromomethane	0.78	0.68	0.12	0.10	0.722
124-72-1	teflurane	-0.07	0.21	0.20	0.00	0.716
126-99-8	2-chloro-1,3-butadiene	0.45	0.36	0.00	0.15	0.709
127-18-4	tetrachloroethylene	0.64	0.44	0.00	0.00	0.837
1333-74-0	hydrogen	0.00	0.00	0.00	0.00	0.109
13838-16-9	enflurane	-0.24	0.40	0.07	0.13	0.901
141-78-6	ethyl acetate	0.11	0.62	0.00	0.45	0.747
142-82-5	heptane	0.00	0.00	0.00	0.00	1.095
151-67-7	halothane	0.10	0.39	0.13	0.05	0.801
156-59-2	cis-1,2-dichloroethene	0.44	0.61	0.11	0.05	0.592
156-60-5	trans-1,2-dichloroethene	0.43	0.41	0.09	0.05	0.592
1634-04-4	methyl tert-butyl ether	0.02	0.21	0.00	0.59	0.872
1717-00-6	1,1-dichloro-1-fluoroethane	0.08	0.43	0.01	0.05	0.673
2234-75-5	1,2,4-trimethylcyclohexane	0.36	0.21	0.00	0.00	1.268
2551-62-4	sulfur hexafluoride	-0.60	-0.20	0.00	0.00	0.584
26675-46-7	isoflurane	-0.24	0.56	0.00	0.08	0.901
2825-83-4	tricyclo[5.2.1.02,6]decane	0.59	0.45	0.00	0.06	1.192
2837-89-0	1-chloro-1,2,2,2-tetrafluoroethane	-0.30	0.17	0.10	0.07	0.666
28523-86-6	sevoflurane	-0.47	0.56	0.00	0.10	0.995
287-92-3	cyclopentane	0.26	0.10	0.00	0.00	0.705
291-64-5	cycloheptane	0.35	0.10	0.00	0.00	0.986
300-57-2	allylbenzene	0.72	0.60	0.00	0.22	1.096
306-83-2	2,2-dichloro-1,1,1-trifluoroethane	-0.16	0.40	0.22	0.00	0.748
3178-22-1	t-butylcyclohexane	0.30	0.10	0.00	0.10	1.409
3221-61-2	2-methyloctane	0.00	0.00	0.00	0.00	1.518

871-83-0	2-methylnonane	0.00	0.00	0.00	0.00	1.518
353-36-6	fluoroethane	0.05	0.35	0.00	0.10	0.428
359-10-4	1-chloro-2,2-difluoroethene	-0.34	0.28	0.15	0.00	0.545
359-35-3	1,1,2,2-tetrafluoroethane	-0.39	0.24	0.10	0.12	0.541
363-72-4	pentafluorobenzene	0.15	0.68	0.00	0.02	0.905
367-11-3	1,2-difluorobenzene	0.39	0.63	0.00	0.06	0.792
367-23-7	1,2,4-trifluorobenzene	0.31	0.65	0.00	0.05	0.830
372-38-3	1,3,5-trifluorobenzene	0.27	0.37	0.00	0.08	0.830
375-17-7	1,1,1,2,2,3,3,4,4-nonafluorobutane	-0.89	-0.43	0.04	0.10	1.012
377-36-6	1,1,2,2,3,3,4,4-octafluorobutane	-0.79	0.08	0.15	0.15	0.974
392-56-3	hexafluorobenzene	0.09	0.56	0.00	0.01	0.943
406-90-6	fluoroxene	0.18	0.30	0.00	0.27	0.801
420-26-8	2-fluoropropane	0.00	0.32	0.00	0.10	0.569
460-13-9	1-fluoropropane	0.03	0.35	0.00	0.13	0.569
4538-51-6	cyanoethylene oxide	0.39	1.00	0.00	0.52	0.495
462-06-6	fluorobenzene	0.48	0.57	0.00	0.10	0.754
526-73-8	1,2,3-trimethylbenzene	0.73	0.61	0.00	0.19	1.139
540-36-3	1,4-difluorobenzene	0.38	0.60	0.00	0.06	0.792
540-54-5	1-chloropropane	0.22	0.40	0.00	0.10	0.654
540-84-1	2,2,4-trimethylpentane	0.00	0.00	0.00	0.00	1.236
541-73-1	1,3-dichlorobenzene	0.85	0.73	0.00	0.02	0.961
543-59-9	1-chloropentane	0.21	0.40	0.00	0.10	0.936
56-23-5	carbon tetrachloride	0.46	0.38	0.00	0.00	0.739
565-61-7	3-methylpentan-2-one	0.11	0.65	0.00	0.51	0.970
565-75-3	2,3,4-trimethylpentane	0.00	0.00	0.00	0.00	1.236
57041-67-5	desflurane	-0.47	0.38	0.05	0.04	0.816
583-57-3	1,2-dimethylcyclohexane	0.32	0.23	0.00	0.00	1.127
583-60-8	2-methylcyclohexanone	0.37	0.83	0.00	0.56	1.002
589-34-4	3-methylhexane	0.00	0.00	0.00	0.00	1.095
591-78-6	2-hexanone	0.14	0.68	0.00	0.51	0.970
592-27-8	2-methylheptane	0.00	0.00	0.00	0.00	1.377
593-60-2	vinyl bromide	0.56	0.50	0.00	0.07	0.522
593-70-4	fluorochloromethane	0.04	0.61	0.07	0.04	0.410
60-29-7	diethyl ether	0.04	0.25	0.00	0.45	0.731
622-97-9	p-methylstyrene	0.87	0.65	0.00	0.18	1.096
628-63-7	pentyl acetate	0.07	0.60	0.00	0.45	1.169
630-08-0	carbon monoxide	0.00	0.00	0.00	0.04	0.222
630-20-6	1,1,1,2-tetrachloroethane	0.54	0.63	0.10	0.08	0.880
637-92-3	ethyl tert-butyl ether	-0.02	0.18	0.00	0.59	1.013
64-17-5	ethanol	0.25	0.42	0.37	0.48	0.449
67-56-1	methanol	0.28	0.44	0.43	0.47	0.308
67-63-0	2-propanol	0.21	0.36	0.33	0.56	0.590
67-64-1	propanone	0.18	0.70	0.04	0.49	0.547
67-66-3	chloroform	0.43	0.49	0.15	0.02	0.617
67-72-1	hexachloroethane	0.68	0.68	0.00	0.00	1.125
679-84-5	halopropane	-0.07	0.28	0.20	0.00	0.851
679-86-7	1,1,2,2,3-pentafluoropropane	-0.45	0.17	0.00	0.03	0.720
71-23-8	1-propanol	0.24	0.42	0.37	0.48	0.590
71-36-3	1-butanol	0.22	0.42	0.37	0.48	0.731
71-41-0	1-pentanol	0.22	0.42	0.37	0.48	0.872

71-43-2	benzene	0.61	0.52	0.00	0.14	0.716
71-55-6	1,1,1-trichloroethane	0.37	0.41	0.00	0.09	0.758
7439-90-9	krypton	0.00	0.00	0.00	0.00	0.246
7440-37-1	argon	0.00	0.00	0.00	0.00	0.190
7440-59-7	helium	0.00	0.00	0.00	0.00	0.068
7440-63-3	xenon	0.00	0.00	0.00	0.00	0.329
74-82-8	methane	0.00	0.00	0.00	0.00	0.250
74-84-0	ethane	0.00	0.00	0.00	0.00	0.390
74-85-1	Ethene	0.11	0.10	0.00	0.07	0.347
74-86-2	acetylene	0.19	0.25	0.21	0.15	0.304
74-87-3	methyl chloride	0.25	0.43	0.00	0.08	0.372
74-95-3	dibromomethane	0.71	0.67	0.10	0.10	0.600
74-97-5	bromochloromethane	0.54	0.80	0.01	0.06	0.547
75-00-3	chloroethane	0.23	0.40	0.00	0.10	0.513
75-01-4	vinyl chloride	0.26	0.38	0.00	0.05	0.470
75-03-6	iodoethane	0.64	0.40	0.00	0.15	0.649
75-09-2	dichloromethane	0.39	0.57	0.10	0.05	0.494
75-10-5	difluoromethane	-0.32	0.49	0.06	0.05	0.325
75-15-0	carbon disulfide	0.88	0.21	0.00	0.07	0.491
75-19-4	cyclopropane	0.18	0.15	0.00	0.00	0.423
75-21-8	ethylene oxide	0.25	0.74	0.07	0.32	0.341
75-25-2	tribromomethane	0.97	0.68	0.15	0.06	0.775
75-26-3	isopropyl bromide	0.33	0.35	0.00	0.14	0.706
75-27-4	bromodichloromethane	0.54	0.66	0.13	0.07	0.669
75-29-6	2-chloropropane	0.18	0.35	0.00	0.12	0.654
75-34-3	1,1-dichloroethane	0.32	0.49	0.10	0.10	0.635
75-35-4	1,1-dichloroethylene	0.36	0.34	0.00	0.05	0.592
75-37-6	1,1-difluoroethane	-0.25	0.49	0.04	0.05	0.466
75-38-7	1,1-difluoroethene	-0.10	0.00	0.00	0.05	0.423
75-56-9	propylene oxide	0.24	0.57	0.00	0.45	0.481
75-65-0	2-methyl-2-propanol	0.28	0.30	0.31	0.60	0.731
75-69-4	fluorotrichloromethane	0.21	0.24	0.00	0.07	0.654
75-71-8	difluorodichloromethane	0.04	0.04	0.00	0.04	0.570
75-73-0	carbon tetrafluoride	-0.58	-0.26	0.00	0.00	0.400
75-83-2	2,2-dimethylbutane	0.00	0.00	0.00	0.00	0.954
75-85-4	tert-amyl alcohol	0.19	0.30	0.31	0.60	0.872
75-88-7	2-chloro-1,1,1-trifluoroethane	0.01	0.40	0.15	0.00	0.626
75995-72-1	1,1,1,2,3,4,4,4-octafluorobutane	-0.79	0.20	0.13	0.05	0.974
76-01-7	pentachloroethane	0.65	0.66	0.17	0.06	1.002
76-14-2	1,2-dichlorotetrafluoroethane	-0.19	0.05	0.00	0.00	0.786
76-38-0	methoxyflurane	0.11	0.67	0.07	0.14	0.910
771-56-2	2,3,4,5,6-pentafluorotoluene	0.16	0.59	0.00	0.01	1.046
7727-37-9	nitrogen	0.00	0.00	0.00	0.00	0.222
7782-44-7	oxygen	0.00	0.00	0.00	0.00	0.183
78-59-1	isophorone	0.51	1.12	0.00	0.53	1.241
78-79-5	2-methyl-1,3-butadiene	0.31	0.23	0.00	0.10	0.727
78-83-1	2-methyl-1-propanol	0.22	0.39	0.37	0.48	0.731
78-87-5	1,2-dichloropropane	0.37	0.60	0.10	0.11	0.776
78-93-3	butan-2-one	0.17	0.70	0.00	0.51	0.688
79-00-5	1,1,2-trichloroethane	0.50	0.68	0.13	0.08	0.758

79-01-6	trichloroethene	0.52	0.37	0.08	0.03	0.715
79-20-9	methyl acetate	0.14	0.64	0.00	0.45	0.606
79-29-8	2,3-dimethylbutane	0.00	0.00	0.00	0.00	0.954
79-34-5	1,1,2,2-tetrachloroethane	0.60	0.76	0.16	0.12	0.880
79-46-9	2-nitropropane	0.22	0.92	0.00	0.32	0.706
811-97-2	1,1,1,2-tetrafluoroethane	-0.41	0.34	0.06	0.04	0.541
871-83-0	2-methylnonane	0.00	0.00	0.00	0.00	1.518
872-05-9	1-decene	0.09	0.08	0.00	0.07	1.475
919-94-8	ethyl tert-pentyl ether	0.00	0.16	0.00	0.61	1.154
930-22-3	1,2-epoxy-3-butene	0.37	0.47	0.00	0.36	0.579
95-47-6	o-xylene	0.66	0.56	0.00	0.16	0.998
95-50-1	1,2-dichlorobenzene	0.87	0.78	0.00	0.04	0.961
95-63-6	1,2,4-trimethylbenzene	0.68	0.56	0.00	0.19	1.139
96-14-0	3-methylpentane	0.00	0.00	0.00	0.00	0.954
96-18-4	1,2,3-trichloropropane	0.55	0.65	0.03	0.31	0.899
96-22-0	3-pentanone	0.15	0.66	0.00	0.51	0.829
96-37-7	methylcyclopentane	0.23	0.10	0.00	0.00	0.845
98-06-6	tert-butylbenzene	0.62	0.49	0.00	0.18	1.280
98-56-6	4-chlorobenzotrifluoride	0.53	0.58	0.00	0.01	1.093
98-82-8	isopropylbenzene	0.60	0.49	0.00	0.16	1.139
994-05-8	tert-amyl methyl ether	0.05	0.21	0.00	0.60	1.013

Table S3. **In-vitro tissue-air** partition coefficients collected by Abraham et al.^{1,3-6,22} and predictions with the PP-LFER-composition-based model.

	adipose		brain		liver		lung		muscle		blood	
	exp	calc	exp	calc	exp	calc	exp	calc	exp	calc	exp	calc
neon		-0.92	-1.81	-1.08		-1.14		-1.18		-1.20	-2.01	-1.17
nitrous oxide	0.04	0.32	-0.31	0.13	-0.38	0.07	-0.37	0.02	-0.43	0.00	-0.34	0.03
ethylbenzene	3.22	3.48		2.55	1.78	2.35		1.91	1.41	1.78	1.47	1.75
styrene	3.52	3.86		2.98	2.14	2.80		2.34	1.67	2.23	1.67	2.22
radon	0.68	-0.02	-0.51	-0.75	-0.51	-0.88		-1.18	-0.81	-1.22	-0.39	-1.18
m-methylstyrene	4.08	4.33		3.39	2.51	3.19		2.74	2.26	2.60	2.28	2.55
propylbenzene		3.90		2.90		2.68		2.25		2.09	1.67	2.01
p-xylene	3.28	3.49		2.56	1.95	2.37		1.93	1.58	1.81	1.61	1.78
1,2-dibromoethane	3.09	3.42		2.84	2.08	2.72		2.30	1.66	2.29	2.08	2.36
propyl bromide	2.37	2.08		1.27	0.91	1.11		0.70	0.62	0.65	0.97	0.68
butane		0.88		-0.06		-0.25		-0.67		-0.79	-0.53	-0.80
1,3-butadiene	1.35	1.39	-0.08	0.60	0.17	0.46	-0.32	0.08	-0.06	0.03	0.09	0.06
1-bromo-2-chloroethane	2.98	2.29		1.71	1.63	1.62		1.27	1.40	1.27	1.60	1.37
allyl chloride	2.00	1.95		1.22		1.09		0.71	1.04	0.67	1.24	0.73
1,2-dichloroethane	2.54	2.47		1.92	1.55	1.81		1.45	1.37	1.44	1.39	1.52
2-methylpentane	1.94	1.76	0.58	0.71	0.65	0.47	-0.10	0.07	0.46	-0.13	-0.39	-0.23
2-pentanone	2.57	2.70		2.34	2.27	2.29		2.20	2.00	2.19	2.14	2.23
1-methoxy-2-propanol		3.70		4.29		4.25		4.29		4.27	4.09	4.29
1-nitropropane	2.70	2.80		2.43	2.18	2.37		2.24	1.46	2.23	2.31	2.29
4-methyl-2-pentanone	2.72	3.04		2.40	2.19	2.32		2.10	1.81	2.10	1.96	2.15
isopropyl acetate	2.48	2.58	1.95	1.99	2.17	1.91		1.72	1.85	1.72	1.55	1.77
m-xylene	3.29	3.51		2.58	1.96	2.39		1.94	1.62	1.82	1.59	1.79
1,3,5-trimethylbenzene		3.99		3.01		2.80		2.36		2.21	1.64	2.14
bromobenzene		3.81		2.96		2.78		2.32		2.21	1.72	2.21
methylcyclohexane	2.65	2.27	1.66	1.22	1.09	0.98		0.57		0.36	0.70	0.23
toluene	2.92	3.04	1.41	2.17	1.64	2.00	1.32	1.55	1.49	1.46	1.14	1.47
chlorobenzene	3.11	3.31		2.46	1.94	2.28		1.83	1.53	1.73	1.63	1.74
2-isopropoxyethanol		3.58		4.03		3.99		4.02		4.00	4.16	4.02
propyl acetate	2.71	2.67	2.00	2.06	2.36	1.97		1.76	1.93	1.76	1.88	1.81
pentane	1.60	1.32	0.34	0.32	0.32	0.10	-0.30	-0.31	-0.15	-0.47	-0.29	-0.53
1-chlorobutane		2.15		1.27		1.11		0.69		0.61	0.63	0.63
2-methoxyethanol		3.72		4.38		4.35		4.40		4.37	4.52	4.39
divinyl ether	1.60	1.74	0.54	0.98	0.48	0.84		0.48	0.34	0.44	0.41	0.49
ethyl formate		1.98		1.84		1.79		1.76		1.74	1.65	1.77
furan	1.81	1.75		1.15	0.77	1.04		0.77		0.75	0.82	0.81
isobutyl acetate	3.05	3.01	2.14	2.24	2.42	2.13		1.81	2.04	1.82	1.69	1.86
2-heptanone	3.66	3.53		2.74	2.63	2.63		2.30	2.33	2.30	2.33	2.35
hexane	2.12	1.76	0.89	0.71	0.69	0.47	0.00	0.07	0.58	-0.13	0.21	-0.23
2-ethoxyethanol		3.62		4.20		4.17		4.21		4.18	4.34	4.20
cyclohexane	2.44	1.96		0.97	1.05	0.75	0.43	0.32	0.52	0.15	0.17	0.08
1-hexanol		3.66		3.36		3.30		2.94		3.00	3.21	3.07
octane	2.72	2.64	1.37	1.52	1.41	1.24		0.87	0.93	0.61	0.55	0.40
1-octene	2.89	2.76	1.81	1.67	1.35	1.41		1.02		0.79	1.07	0.63
2-butoxyethanol		3.93		3.99		3.96		3.90		3.90	3.90	3.93
nonane	2.92	3.09	1.69	1.93		1.64		1.29		1.00	1.17	0.74
undecane		3.97		2.77		2.46		2.13		1.81	1.31	1.46

dodecane		4.41		3.20		2.88		2.56		2.23	1.39	1.84
propene	0.76	0.57	-0.27	-0.18	-0.28	-0.32	-0.24	-0.64	-0.27	-0.69	-0.21	-0.65
dimethyl ether		0.79		0.94		0.90		0.92		0.89	1.16	0.91
1,1,2,4,4-pentafluorobutane		2.02		1.45		1.37		1.01		1.03	0.87	1.15
3-methyl-1-butanol	3.18	3.17	2.79	3.04	2.97	3.00		2.76	2.90	2.79	2.75	2.85
butyl acetate	3.18	3.08	2.22	2.29	2.45	2.18		1.85	2.20	1.85	1.94	1.90
1,4-dioxane		2.80		3.12		3.08		3.11		3.09	3.27	3.11
isopentyl acetate	3.44	3.45	2.34	2.56	2.55	2.42		2.03	2.32	2.02	1.79	2.05
1-nonene	3.24	3.20	2.04	2.07	1.71	1.80		1.43		1.17	1.18	0.96
decane	3.05	3.53		2.35	2.09	2.05		1.71		1.40	1.17	1.09
chlorodibromomethane	3.28	3.30		2.69	2.10	2.57		2.11	1.75	2.08	1.88	2.14
teflurane	1.30	1.62	0.05	0.94	0.01	0.81		0.36	0.34	0.33	-0.22	0.41
2-chloro-1,3-butadiene		2.14		1.33		1.18	1.12	0.77		0.71	0.77	0.73
tetrachloroethylene	3.21	2.87		1.95	1.85	1.75	1.86	1.30	1.30	1.16	1.19	1.12
hydrogen		-0.85		-1.08		-1.14		-1.19	-1.76	-1.22	-1.77	-1.19
enflurane	1.97	2.04	0.49	1.20	0.62	1.05	0.56	0.64	0.64	0.60	0.35	0.67
ethyl acetate	2.18	2.29	1.90	1.97	2.03	1.91		1.83	1.84	1.82	1.90	1.86
heptane	2.51	2.20	0.94	1.11	0.96	0.85	0.40	0.46	0.85	0.23	0.50	0.08
halothane	2.27	2.25	0.71	1.50	0.86	1.36	0.46	0.91	0.74	0.87	0.57	0.93
cis-1,2-dichloroethene	2.36	2.32		1.77	1.18	1.66		1.27	0.78	1.25	1.16	1.34
trans-1,2-dichloroethene	2.17	1.97		1.32	0.95	1.19		0.78	0.55	0.75	0.88	0.81
methyl tert-butyl ether	1.97	1.98	1.54	1.44	1.35	1.37		1.24	1.57	1.24	1.18	1.26
1,1-dichloro-1-fluoroethane	1.56	1.66		0.86	0.52	0.72		0.32	0.02	0.27	0.32	0.33
1,2,4-trimethylcyclohexane	3.19	3.52	1.89	2.41	1.54	2.14		1.76		1.50	0.87	1.30
sulfur hexafluoride		-0.47	-1.78	-1.39		-1.56	-2.14	-1.92	-1.85	-2.00	-2.17	-1.97
isoflurane	1.92	2.14	0.39	1.24	0.56	1.08	0.20	0.68	0.31	0.61	0.20	0.68
tricyclo[5.2.1.02,6]decane	4.01	3.95		2.91	2.74	2.67		2.25		2.04	1.76	1.91
1-chloro-1,2,2,2-tetrafluoroethane		0.93		0.19		0.07		-0.29		-0.32	0.18	-0.25
sevoflurane	1.62	2.15	0.10	1.21	0.43	1.04	0.11	0.64	0.22	0.57	-0.20	0.64
cyclopentane		1.47		0.53		0.34		-0.09		-0.22	0.24	-0.25
cycloheptane		2.46		1.42		1.18		0.77		0.56	0.72	0.43
allylbenzene		4.07		3.13		2.94		2.49		2.37	1.71	2.33
2,2-dichloro-1,1,1-trifluoroethane	1.80	1.93		1.31	0.67	1.20		0.77	0.39	0.75	0.61	0.86
t-butylcyclohexane	3.17	3.74	1.77	2.60	1.68	2.32		1.96		1.68	1.16	1.44
2-methyloctane	2.56	3.53	1.50	2.35	1.17	2.05		1.71		1.40	0.52	1.09
2-methylnonane	2.84	3.53		2.35		2.05		1.71		1.40	0.76	1.09
fluoroethane		0.74		0.22		0.14		-0.03		-0.05	0.09	0.00
1-chloro-2,2-difluoroethene	1.30	0.75	0.04	0.18	0.04	0.09	0.00	-0.22	0.00	-0.23	0.06	-0.13
1,1,2,2-tetrafluoroethane		0.55		0.06		-0.02		-0.21		-0.22	-0.12	-0.15
pentafluorobenzene		2.82		1.92		1.75		1.32		1.22	0.51	1.27
1,2-difluorobenzene		2.71		1.87		1.71		1.28		1.21	0.96	1.25
1,2,4-trifluorobenzene		2.75		1.89		1.73		1.29		1.21	0.76	1.26
1,3,5-trifluorobenzene		2.29		1.38		1.20		0.77		0.67	0.49	0.67
1,1,1,2,2,3,3,4,4-nonafluorobutane		0.25		-0.84		-1.07		-1.46		-1.63	-1.52	-1.70
1,1,2,2,3,3,4,4-octafluorobutane		1.24		0.38		0.24		-0.17		-0.20	-0.36	-0.12
hexafluorobenzene		2.68		1.73		1.54		1.11		0.99	0.39	1.01
fluoroxene	1.53	2.02	0.29	1.20	0.33	1.06		0.70	0.35	0.66	0.15	0.69
2-fluoropropane		1.06		0.34		0.23		-0.07		-0.10	0.06	-0.04
1-fluoropropane		1.15		0.47		0.36		0.09		0.07	0.02	0.13
cycanoethylene oxide	3.11	2.90		3.43	2.66	3.40		3.45		3.42	3.22	3.44

fluorobenzene	2.62		1.81		1.66		1.23		1.16	1.06	1.20
1,2,3-trimethylbenzene	4.23		3.26		3.06		2.62		2.47	1.82	2.41
1,4-difluorobenzene	2.66		1.81		1.65		1.22		1.14	0.87	1.18
1-chloropropane	2.07	1.72	0.94	0.71	0.79		0.42	0.32	0.37	0.59	0.42
2,2,4-trimethylpentane	2.47	2.64	1.52	1.03	1.24		0.87	0.52	0.61	0.23	0.40
1,3-dichlorobenzene		3.97	3.06		2.86		2.40		2.26	2.30	2.23
1-chloropentane		2.59	1.64		1.45		1.02		0.90	0.87	0.90
carbon tetrachloride	2.56	2.24	1.36	1.15	1.17		0.73	0.66	0.61	0.62	0.61
3-methylpentan-2-one		3.04	2.40		2.32		2.10		2.10	2.23	2.15
2,3,4-trimethylpentane	2.65	2.64	1.52	1.27	1.24		0.87	0.64	0.61	0.57	0.40
desflurane		1.39	0.54		0.39		0.00		-0.05	-0.37	0.04
1,2-dimethylcyclohexane	3.24	3.06	1.98	1.99	1.74	1.74	1.34		1.11	0.91	0.97
2-methylcyclohexanone		3.76		3.18		3.11	2.89		2.90	2.87	2.95
3-methylhexane	2.44	2.20	1.01	1.11	1.04	0.85	0.46	1.04	0.23	0.11	0.08
2-hexanone		3.11		2.49		2.41	2.19		2.20	2.10	2.24
2-methylheptane	2.30	3.09	1.35	1.93	0.75	1.64	1.29		1.00	0.49	0.74
vinyl bromide	1.69	1.90		1.22	0.52	1.10	0.74	0.35	0.70	0.49	0.75
fluorochloromethane	1.19	1.18		0.85	0.54	0.78	0.63	0.39	0.61	0.71	0.68
diethyl ether	1.74	1.59	1.10	1.09	0.94	1.01	1.14	0.89	0.87	0.88	1.11
p-methylstyrene	4.05	4.33		3.40	2.51	3.20		2.75	2.26	2.61	2.37
pentyl acetate	3.57	3.51	2.38	2.62	2.64	2.47		2.08	2.36	2.06	1.98
carbon monoxide		-0.51		-0.91		-0.99	-1.69	-1.09		-1.12	-1.67
1,1,1,2-tetrachloroethane	3.33	3.37		2.61	1.95	2.46		2.00	1.60	1.93	1.55
ethyl tert-butyl ether	2.13	2.31	1.29	1.54	1.52	1.43		1.17	1.35	1.17	1.07
ethanol	2.34	2.50	3.15	3.08		3.04	3.07	3.08	3.08	3.05	3.27
methanol	2.33	2.76		3.43		3.40	3.24	3.45		3.42	3.41
2-propanol	2.35	2.52	2.91	2.99	2.99	2.96	2.77	2.97	2.87	2.95	3.02
propanone	1.91	2.18	2.19	2.53	2.38	2.49	2.20	2.52	2.21	2.50	2.36
chloroform	2.38	2.27	1.30	1.69	1.28	1.57	1.12	1.14	1.12	1.12	1.15
hexachloroethane	3.52	4.19		3.19	2.57	2.97		2.54	1.88	2.35	1.76
halopropane	2.35	2.14		1.40		1.26		0.79		0.74	0.75
1,1,2,2,3-pentafluoropropane		0.71		-0.20		-0.37		-0.74		-0.81	-0.48
1-propanol	2.54	2.61	2.98	3.00	3.11	2.97	2.91	2.95	2.95	2.94	3.06
1-butanol	2.95	2.88	3.06	3.01	3.10	2.97		2.87	2.95	2.87	3.08
1-pentanol	3.41	3.25	3.03	3.13	3.24	3.09		2.85	2.91	2.88	2.83
benzene	2.64	2.61	1.25	1.83	1.26	1.68	1.08	1.26	1.11	1.20	1.05
1,1,1-trichloroethane	2.41	2.25	0.92	1.39	1.07	1.23	0.67	0.80	0.67	0.72	0.63
krypton	-0.48	-0.45	-1.38	-0.98	-1.14	-1.07		-1.23	-1.38	-1.26	-1.22
argon		-0.62	-1.49	-1.04		-1.11		-1.23	-1.64	-1.25	-1.52
helium		-0.96		-1.08		-1.14	-1.98	-1.16	-1.93	-1.19	-2.00
xenon	0.26	-0.19	-0.70	-0.85	-1.00	-0.97		-1.21	-1.00	-1.25	-0.85
methane		-0.44	-1.39	-0.97		-1.07		-1.23	-1.41	-1.26	-1.42
ethane		0.00		-0.74		-0.87	-1.28	-1.17		-1.21	-1.02
Ethene	0.20	0.17	-0.23	-0.39	-0.30	-0.49	-0.25	-0.68	-0.20	-0.71	-0.53
acetylene		0.85		0.90		0.85	-0.06	0.79	0.03	0.77	-0.06
methyl chloride	1.13	0.93		0.48	0.54	0.40		0.24	-0.01	0.22	0.31
dibromomethane	2.90	2.78		2.25	1.83	2.14		1.75	1.61	1.73	1.87
bromochloromethane	2.51	2.41		1.83	1.47	1.72		1.40	1.05	1.38	1.21
chloroethane	1.59	1.29		0.65	0.56	0.54		0.26	0.51	0.24	0.49
vinyl chloride	1.30	1.16		0.50	0.20	0.38		0.09	0.32	0.06	0.17

iodoethane		2.26		1.50		1.35		0.94		0.89	0.83	0.92
dichloromethane	2.01	1.88	0.78	1.40	1.01	1.30	0.76	0.99	0.79	0.98	1.12	1.06
difluoromethane	0.16	0.29		0.20	0.44	0.15		0.11	0.16	0.09	0.20	0.13
carbon disulfide		1.77	0.90	1.01	0.78	0.86		0.43		0.36	0.30	0.36
cyclopropane	0.99	0.55	-0.10	-0.18	-0.19	-0.31	-0.51	-0.64	-0.31	-0.69	-0.21	-0.65
ethylene oxide	1.64	1.79	1.81	2.23	1.73	2.20	1.79	2.23	1.75	2.21	1.80	2.23
tribromomethane		3.76		3.15		3.01		2.51		2.46	2.15	2.50
isopropyl bromide	2.20	1.96		1.15	0.64	1.00		0.60	0.61	0.54	0.64	0.57
bromodichloromethane	2.72	2.81		2.25	1.49	2.13		1.71		1.69	1.49	1.77
2-chloropropane	1.84	1.60		0.81	0.50	0.67		0.31	0.31	0.27	0.32	0.31
1,1-dichloroethane	2.21	2.12		1.51	1.03	1.40		1.02	0.71	1.00	0.88	1.08
1,1-dichloroethylene	1.84	1.61		0.82	0.65	0.67		0.27	0.31	0.21	0.70	0.25
1,1-difluoroethane		0.74		0.28		0.21		0.05		0.03	0.42	0.10
1,1-difluoroethene	-0.10	-0.02		-0.73	-0.10	-0.85		-1.11	-0.54	-1.15	-0.74	-1.11
propylene oxide		1.72		1.98		1.94		1.97		1.95	1.80	1.97
2-methyl-2-propanol	2.20	2.72	2.79	2.93	2.69	2.89		2.85	2.80	2.84	2.68	2.87
fluorotrichloromethane		1.46		0.62		0.46		0.05		-0.02	-0.06	0.01
difluorodichloromethane		0.67		-0.18		-0.34		-0.73		-0.80	-0.82	-0.77
carbon tetrafluoride		-1.11		-1.89		-2.01		-2.26		-2.30	-1.10	-2.26
2,2-dimethylbutane	1.82	1.76	0.45	0.71	0.54	0.47	-0.22	0.07	0.00	-0.13	-0.59	-0.23
tert-amyl alcohol	2.66	2.97	2.66	2.90	2.68	2.86		2.73	2.59	2.74	2.59	2.78
2-chloro-1,1,1-trifluoroethane	1.43	1.63	0.26	1.01	0.31	0.90	0.34	0.49	0.22	0.47	0.14	0.57
1,1,1,2,3,4,4,4-octafluorobutane		1.36		0.46		0.31		-0.10		-0.16	-0.59	-0.08
pentachloroethane	3.61	4.07		3.32	2.41	3.17		2.68	1.86	2.60	2.02	2.63
1,2-dichlorotetrafluoroethane		1.06		0.08		-0.12		-0.53		-0.66	-0.82	-0.68
methoxyflurane	2.90	2.93	1.40	2.15	1.47	2.01	1.20	1.59	1.35	1.56	1.28	1.64
2,3,4,5,6-pentafluorotoluene		3.14		2.15		1.95		1.52		1.37	0.73	1.36
nitrogen	-1.27	-0.52	-1.80	-1.00		-1.09		-1.23		-1.26	-1.83	-1.22
oxygen		-0.64		-1.04		-1.12	-1.62	-1.22		-1.25	-1.58	-1.22
isophorone		5.10		4.34		4.22		3.84		3.85	3.37	3.92
2-methyl-1,3-butadiene	1.84	1.82		0.94	0.42	0.77		0.34	0.26	0.25	0.10	0.26
2-methyl-1-propanol	2.73	2.82	2.78	2.93	2.94	2.90	2.60	2.79	2.74	2.79	2.92	2.84
1,2-dichloropropane	2.70	2.79		2.11	1.39	1.98		1.55	1.08	1.52	1.14	1.59
butan-2-one	2.26	2.38	2.07	2.39	2.36	2.34	2.01	2.34	2.14	2.32	2.24	2.35
1,1,2-trichloroethane	3.16	3.07		2.44	1.86	2.32		1.88	1.36	1.85	1.67	1.93
trichloroethene	2.75	2.39	1.25	1.62	1.41	1.46	1.15	1.00	1.13	0.93	1.14	0.95
methyl acetate	2.00	1.99	1.85	2.03	1.95	1.99		1.99	1.81	1.97	1.98	1.99
2,3-dimethylbutane		1.76		0.71		0.47		0.07		-0.13	0.78	-0.23
1,1,2,2-tetrachloroethane	3.58	3.76		3.13	2.29	3.01		2.54	2.00	2.52	2.13	2.59
2-nitropropane	2.19	2.73		2.36	1.80	2.31		2.18	1.46	2.17	2.23	2.23
1,1,1,2-tetrafluoroethane		0.58		-0.03		-0.12		-0.36		-0.38	-0.25	-0.30
2-methylnonane		3.53	1.81	2.35	1.45	2.05		1.71		1.40		1.09
1-decene	3.36	3.65	2.17	2.49	1.98	2.20		1.84		1.56	1.21	1.30
ethyl tert-pentyl ether		2.75		1.85		1.70		1.35		1.34	1.25	1.34
1,2-epoxy-3-butene	2.23	1.87		1.53	1.74	1.47		1.37	1.66	1.36	1.97	1.39
o-xylene	3.20	3.62	1.39	2.70	1.76	2.51	1.94	2.06	1.71	1.94	1.42	1.92
1,2-dichlorobenzene		4.08		3.18		2.99		2.54		2.40	2.63	2.38
1,2,4-trimethylbenzene	3.25	4.09	1.63	3.11	2.20	2.91		2.47		2.32	1.85	2.26
3-methylpentane	2.01	1.76	0.64	0.71	0.69	0.47	-0.05	0.07	0.58	-0.13	-0.37	-0.23
1,2,3-trichloropropane		3.39		2.64		2.51		2.11		2.09	2.01	2.14

3-pentanone		2.68		2.31		2.25		2.15		2.15	2.21	2.19
methylcyclopentane	2.25	1.86	0.86	0.87	0.89	0.65	0.23	0.22	0.70	0.06	-0.07	-0.01
tert-butylbenzene	3.26	4.35	1.67	3.31	2.32	3.07		2.66		2.46	1.24	2.34
4-chlorobenzotrifluoride	2.99	3.75	1.60	2.74	1.71	2.52	1.63	2.10		1.91	1.43	1.84
isopropylbenzene		3.88		2.89		2.67		2.24		2.07	1.57	2.00
tert-amyl methyl ether	2.52	2.45	1.39	1.69	1.50	1.58		1.33	1.38	1.32	1.22	1.35

Table S4. Skin-water partition coefficients from refs 12 and 23 and predictions with the PP-LFER-composition-based model.

	<i>E</i>	<i>S</i>	<i>A</i>	<i>B</i>	<i>V</i>	exp	calc
water	0.00	0.45	0.82	0.35	0.167	-0.29	-0.20
methanol	0.28	0.44	0.43	0.47	0.308	-0.22	-0.19
ethanol	0.25	0.42	0.37	0.48	0.449	-0.13	-0.16
1-propanol	0.24	0.42	0.37	0.48	0.590	0.21	-0.09
1-butanol	0.22	0.42	0.37	0.48	0.731	0.40	0.07
1-pentanol	0.22	0.42	0.37	0.48	0.872	0.70	0.34
1-hexanol	0.21	0.42	0.37	0.48	1.013	1.00	0.71
1-heptanol	0.21	0.42	0.37	0.48	1.154	1.48	1.16
1-octanol	0.20	0.42	0.37	0.48	1.295	1.72	1.62
1-decanol	0.19	0.42	0.37	0.48	1.576	2.28	2.63
benzene	0.61	0.52	0.00	0.14	0.716	1.48	1.28
o-phenylenediamine	1.26	1.40	0.24	0.73	0.916	0.84	-0.03
2-Nitro-p-phenylenediamine	1.53	2.05	0.35	0.70	1.090	1.11	0.17
Lidocaine	1.10	1.47	0.06	1.24	2.059	1.88	1.55
4-Methylphenol	0.82	0.87	0.57	0.31	0.916	1.34	0.97
4-Amino-2-nitrophenol	1.36	1.50	0.30	0.66	1.049	1.11	0.31
4-Hydroxybenzyl alcohol	1.00	1.15	0.88	0.85	0.975	0.95	-0.05
4-Hydroxy-phenylacetic acid	1.03	1.44	0.94	0.74	1.131	1.15	0.28
4-Hydroxy-methylphenylacetate	0.91	1.46	0.59	0.68	1.272	1.11	0.65
4-Hydroxyphenylacetamide	1.28	2.03	0.86	0.94	1.172	0.70	-0.05
8-Methoxypsoralen	1.61	1.70	0.00	0.80	1.450	1.47	1.15
Progesterone	1.45	3.29	0.00	1.14	2.622	2.52	2.71
Testosterone	1.54	2.59	0.32	1.19	2.383	2.03	2.14
Corticosterone	1.86	3.43	0.40	1.63	2.739	1.59	1.56
Hydrocortisone	2.03	3.49	0.71	1.90	2.798	0.85	1.05
Hydrocortisone-21-acetate	1.89	2.88	0.46	2.16	3.095	1.23	1.34
Hydrocortisone-21-propanoate(1h)	1.87	2.90	0.46	2.16	3.236	1.51	1.75
Hydrocortisone-21-pentanoate	1.83	2.98	0.46	2.16	3.518	2.10	2.56
Hydrocortisone-21-hexanoate(1j)	1.81	3.02	0.46	2.16	3.659	2.39	2.97
Hydrocortisone-21-octanoate(1k)	1.77	3.05	0.46	2.16	3.941	2.91	3.85
Cortisone	1.96	3.50	0.36	1.87	2.755	0.91	0.90
Cortisone-21-acetate	1.82	3.11	0.21	2.13	3.052	1.30	1.12
Cortisone-21-butanoate	1.78	3.45	0.21	2.13	3.334	1.80	1.77
Cortisone-21-hexanoate	1.74	3.86	0.21	2.13	3.616	2.43	2.40
Cortisone-21-octanoate	1.70	4.00	0.21	2.13	3.898	2.93	3.20
Aldosterone	2.01	3.47	0.40	1.90	2.689	1.20	0.68
Estradiol	1.80	1.77	0.86	1.10	2.199	1.93	2.49
1c	1.99	3.11	0.46	2.61	3.592	1.34	1.41
1d	2.10	3.15	1.06	2.61	3.451	1.04	1.17
1i	1.93	3.49	0.46	2.61	4.015	2.12	2.47
1e	2.02	3.58	1.06	2.61	3.874	1.83	2.16
1g	2.02	3.49	0.83	2.64	3.717	1.30	1.60
1a	2.31	3.32	1.01	2.84	3.492	0.85	0.65
1b	2.21	3.77	0.46	2.86	3.774	1.08	0.99
1f	2.21	4.15	0.96	2.78	3.915	1.40	1.56
Perchloroethylene	0.64	0.44	0.00	0.00	0.837	1.93	2.37
Trichloroethylene	0.52	0.37	0.08	0.03	0.715	1.52	1.68

Benzene	0.61	0.52	0.00	0.14	0.716	0.91	1.28
Hexane	0.00	0.00	0.00	0.00	0.954	2.20	2.84
Toluene	0.60	0.52	0.00	0.14	0.857	1.15	1.80
m-Xylene	0.62	0.52	0.00	0.16	0.998	1.24	2.28
Styrene	0.85	0.65	0.00	0.16	0.955	1.00	2.14
Methylene chloride	0.39	0.57	0.10	0.05	0.494	0.35	0.58
Carbon tetrachloride	0.46	0.38	0.00	0.00	0.739	1.57	1.93
Methyl chloroform	0.37	0.41	0.00	0.09	0.758	1.10	1.56
Halothane	0.10	0.39	0.13	0.05	0.801	1.04	1.58
Isoflurane	-0.24	0.56	0.00	0.08	0.901	0.87	1.61
Dibromomethane	0.71	0.69	0.11	0.07	0.600	0.52	0.94

See ref 12 for identities of hydrocortisone derivatives (1c, 1d, 1i, 1e, 1g, a1, 1b, 1f)

Table S5. Milk-water partition coefficients from ref 7 and predictions with the PP-LFER-composition-based model.

	<i>E</i>	<i>S</i>	<i>A</i>	<i>B</i>	<i>V</i>	exp	calc
Methyl acetate	0.14	0.64	0.00	0.45	0.610	-0.09	0.00
Ethyl acetate	0.11	0.62	0.00	0.45	0.750	-0.03	0.08
n-Propyl acetate	0.09	0.60	0.00	0.45	0.890	0.10	0.27
n-Butyl acetate	0.07	0.60	0.00	0.45	1.030	0.39	0.62
n-Pentyl acetate	0.07	0.60	0.00	0.45	1.170	0.93	1.09
Ethanol	0.25	0.42	0.37	0.48	0.450	-0.04	-0.03
Propan-2-ol	0.21	0.36	0.33	0.56	0.590	-0.04	-0.02
Butan-1-ol	0.22	0.42	0.37	0.48	0.730	-0.01	0.04
Pentan-1-ol	0.22	0.42	0.37	0.48	0.870	0.05	0.16
Heptan-1-ol	0.21	0.42	0.37	0.48	1.150	0.58	0.78
Octan-1-ol	0.20	0.42	0.37	0.48	1.290	1.09	1.23
Nonan-1-ol	0.19	0.42	0.37	0.48	1.440	1.66	1.77
n-Pentane	0.00	0.00	0.00	0.00	0.810	2.33	1.99
n-Hexane	0.00	0.00	0.00	0.00	0.950	2.91	2.55
n-Heptane	0.00	0.00	0.00	0.00	1.090	3.58	3.12
n-Octane	0.00	0.00	0.00	0.00	1.240	4.13	3.73
n-Nonane	0.00	0.00	0.00	0.00	1.380	4.73	4.30
n-Decane	0.00	0.00	0.00	0.00	1.520	5.34	4.88
Dichloromethane	0.39	0.57	0.10	0.05	0.490	0.00	0.00
Trichloromethane	0.43	0.49	0.15	0.02	0.620	0.73	0.81
Tetrachloromethane	0.46	0.38	0.00	0.00	0.740	1.88	1.63
1,1,2,2-Tetrachloroethane	0.60	0.76	0.16	0.12	0.880	1.24	1.23
1-Chloropentane	0.21	0.40	0.00	0.10	0.940	2.08	1.84
1-Chloroheptane	0.19	0.40	0.00	0.10	1.220	3.17	2.95
1-Chlorooctane	0.19	0.40	0.00	0.09	1.360	3.89	3.56
Butanal	0.19	0.65	0.00	0.45	0.690	-0.04	0.04
Hexanal	0.15	0.65	0.00	0.45	0.970	0.65	0.45
Heptanal	0.14	0.65	0.00	0.45	1.110	1.13	0.87
Octanal	0.16	0.65	0.00	0.45	1.250	1.67	1.39
Nonanal	0.15	0.65	0.00	0.45	1.390	2.08	1.93
1-Hexene	0.08	0.08	0.00	0.07	0.910	2.21	2.08
1-Heptene	0.09	0.08	0.00	0.07	1.050	2.70	2.65
1-Octene	0.09	0.08	0.00	0.07	1.190	3.29	3.22
1-Nonene	0.09	0.08	0.00	0.07	1.330	3.94	3.79
1-Decene	0.09	0.08	0.00	0.07	1.470	4.49	4.36
Benzene	0.61	0.52	0.00	0.14	0.720	0.93	0.99
Toluene	0.60	0.52	0.00	0.14	0.860	1.41	1.51
Ethylbenzene	0.61	0.51	0.00	0.15	1.000	1.90	2.03
n-Propylbenzene	0.60	0.50	0.00	0.15	1.140	2.44	2.60
n-Butylbenzene	0.60	0.51	0.00	0.15	1.280	2.89	3.15
Chlorobenzene	0.72	0.65	0.00	0.07	0.840	1.62	1.65
1,2-Dichlorobenzene	0.87	0.78	0.00	0.04	0.960	2.20	2.22
1,3-Dichlorobenzene	0.85	0.73	0.00	0.02	0.960	2.34	2.33
1,2,4-Trichlorobenzene	0.98	0.81	0.00	0.00	1.080	2.86	2.90
Di-n-propyl ether	0.01	0.25	0.00	0.45	1.010	0.77	0.80
Di-n-butyl ether	0.00	0.25	0.00	0.45	1.290	1.84	1.84
Dipentyl ether	0.00	0.25	0.00	0.45	1.580	2.88	3.01

2-Butanone	0.17	0.70	0.00	0.51	0.690	-0.04	0.00
2-Pentanone	0.14	0.68	0.00	0.51	0.830	0.03	0.09
2-Hexanone	0.14	0.68	0.00	0.51	0.970	0.19	0.29
2-Heptanone	0.12	0.68	0.00	0.51	1.110	0.58	0.64
2-Octanone	0.11	0.68	0.00	0.51	1.250	1.13	1.11
2-Nonanone	0.12	0.68	0.00	0.51	1.390	1.69	1.65
Nitromethane	0.31	0.95	0.06	0.32	0.420	-0.02	-0.02
Nitroethane	0.27	0.95	0.02	0.33	0.560	0.02	0.01
1-Nitropropane	0.24	0.95	0.00	0.31	0.710	0.11	0.12
1-Nitrobutane	0.23	0.95	0.00	0.29	0.850	0.36	0.41
1-Nitrohexane	0.20	0.95	0.00	0.29	1.130	1.35	1.29
Nitrobenzene	0.87	1.11	0.00	0.28	0.890	0.74	0.75
2-Nitrotoluene	0.87	1.11	0.00	0.28	1.030	1.18	1.23
2,6-Dinitrotoluene	1.15	1.60	0.00	0.45	1.210	1.04	1.01
2,4-Dinitrotoluene	1.17	1.27	0.07	0.51	1.210	1.01	1.01
1-Chloro-4-nitrobenzene	0.98	1.18	0.00	0.24	1.010	1.24	1.31
Anthracene	2.29	1.34	0.00	0.28	1.450	3.53	3.61
Phenanthrene	2.06	1.29	0.00	0.29	1.450	3.50	3.46
Fluoranthene	2.38	1.55	0.00	0.20	1.580	3.90	4.30
Pyrene	2.81	1.71	0.00	0.29	1.580	3.98	4.06
Fluorene	1.59	1.03	0.00	0.20	1.360	3.08	3.41
Cyclopentane	0.26	0.10	0.00	0.00	0.700	2.29	1.63
Cyclohexane	0.31	0.10	0.00	0.00	0.850	2.69	2.26
Cycloheptane	0.35	0.10	0.00	0.00	0.990	3.41	2.85
Cyclooctane	0.41	0.10	0.00	0.00	1.130	4.11	3.46
Cyclohexene	0.40	0.20	0.00	0.09	0.800	2.01	1.66
Methylcyclohexane	0.24	0.06	0.00	0.00	0.990	3.17	2.82
Methylcyclopentane	0.23	0.10	0.00	0.00	0.850	3.10	2.20
Ethyl tert-butyl ether	-0.02	0.16	0.00	0.60	1.010	0.30	0.40
Ethyl tert-pentyl ether	0.00	0.16	0.00	0.61	1.150	0.74	0.80
Diisopropyl ether	-0.06	0.16	0.00	0.58	1.010	0.08	0.44
Halothane	0.10	0.39	0.13	0.05	0.740	1.07	1.03
Enflurane	-0.24	0.54	0.01	0.10	0.800	0.75	0.88
Isoflurane	-0.24	0.56	0.00	0.08	0.800	0.83	0.94
Methoxyflurane	0.11	0.67	0.07	0.14	0.870	0.93	1.00
Hexafluorobenzene	0.09	0.56	0.00	0.01	0.820	1.11	1.48
Methylpentafluorobenzene	0.06	0.59	0.00	0.01	0.950	1.79	1.94
Pentafluorobenzene	0.15	0.68	0.00	0.02	0.800	0.97	1.29
2-Ethyl-1-hexanol	0.21	0.39	0.37	0.48	1.290	0.94	1.26
3-Ethyl-3-hexanol	0.20	0.30	0.31	0.60	1.290	0.59	0.97
4-Ethyl-3-hexanol	0.17	0.36	0.33	0.57	1.290	0.77	0.99
3-Ethyl-3-pentanol	0.20	0.30	0.31	0.60	1.150	0.18	0.55
Tetrachloroethene	0.64	0.44	0.00	0.00	0.840	2.26	2.09
Trichloroethene	0.52	0.37	0.08	0.03	0.710	1.51	1.34
3-Chlorophenol	0.91	1.06	0.69	0.15	0.900	0.67	0.87
4-Chlorophenol	0.92	1.08	0.67	0.20	0.900	0.57	0.73
4-Ethylpyridine	0.63	0.80	0.00	0.57	0.960	0.16	0.26
2-Chloroaniline	1.03	0.92	0.25	0.31	0.940	0.81	0.82
Acenaphthene	1.60	1.04	0.00	0.20	1.260	2.65	3.00
4-Nitroanisole	0.97	1.29	0.00	0.40	1.090	1.00	0.92

1,4-Dimethoxybenzene	0.81	1.00	0.00	0.50	1.120	0.81	0.83
Ethyl benzoate	0.69	0.85	0.00	0.46	1.210	1.25	1.36
Indole	1.20	1.12	0.44	0.22	0.950	0.82	0.98
Biphenyl	1.36	0.99	0.00	0.26	1.320	2.82	2.89
1-Naphthol	1.52	1.05	0.61	0.37	1.140	1.28	1.26
2,2,4-Trimethylpentane	0.00	0.00	0.00	0.00	1.240	3.02	3.73
4-Bromophenol	1.08	1.17	0.67	0.20	0.950	0.76	0.90
4-Iodoaniline	1.53	1.28	0.31	0.40	1.070	0.92	0.92
N,N-Diethylaniline	0.95	0.80	0.00	0.50	1.380	1.32	2.08
4-n-Propylphenol	0.79	0.88	0.55	0.37	1.200	1.08	1.19
4-Iodophenol	1.38	1.22	0.68	0.20	1.030	1.08	1.25

Table S6. **Tissue-blood**, **tissue-plasma**, and **tissue-serum** steady-state concentration ratios of neutral chemicals measured in vivo and the corresponding equilibrium partition coefficients predicted by the PP-LFER-composition model. Measured values are from data compilations in refs 1, 2, 4-6, 24.

		<i>E</i>	<i>S</i>	<i>A</i>	<i>B</i>	<i>V</i>	species	blood/plasma/ serum	measured	predicted
<i>adipose</i>										
80-07-9	p,p'-dichlorodiphenyl sulfone	1.88	2.20	0.00	0.54	1.850	rat	blood	2.03	1.55
2051-62-9	PCB 3	1.50	1.05	0.00	0.18	1.447	rat	blood	1.48	2.02
2050-68-2	PCB 15	1.65	1.18	0.00	0.16	1.569	rat	blood	1.85	2.11
37680-73-2	PCB 101	2.04	1.61	0.00	0.13	1.936	rat	blood	1.85	2.28
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	rat	blood	2.60	2.35
33284-52-5	PCB 80	1.96	1.44	0.00	0.11	1.814	rat	blood	2.34	2.26
35694-04-3	PCB 133	2.20	1.74	0.00	0.11	2.059	rat	blood	2.01	2.35
38411-22-2	PCB 136	2.14	1.74	0.00	0.11	2.059	rat	blood	1.95	2.34
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	rat	blood	2.06	2.35
33979-03-2	PCB 155	2.12	1.74	0.00	0.11	2.059	rat	blood	1.95	2.34
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	1.13	1.80
938-73-8	ethenzamide	0.91	1.51	0.49	0.80	1.313	rat	plasma	-0.17	-0.07
2050-68-2	PCB 15	1.65	1.18	0.00	0.16	1.569	human	plasma	1.95	2.26
7012-37-5	PCB 28	1.76	1.33	0.00	0.15	1.691	human	plasma	2.30	2.31
35693-99-3	PCB 52	1.90	1.48	0.00	0.15	1.814	human	plasma	1.90	2.36
70362-47-9	PCB 48	1.89	1.48	0.00	0.15	1.814	human	plasma	1.95	2.36
2437-79-8	PCB 47	1.88	1.48	0.00	0.15	1.814	human	plasma	2.49	2.35
41464-39-5	PCB 44	1.90	1.48	0.00	0.15	1.814	human	plasma	1.90	2.36
32690-93-0	PCB 74	1.91	1.46	0.00	0.13	1.814	human	plasma	2.30	2.38
32598-10-0	PCB 66	1.91	1.46	0.00	0.13	1.814	human	plasma	2.20	2.38
32598-11-1	PCB 70	1.89	1.46	0.00	0.13	1.814	human	plasma	1.85	2.37
38379-99-6	PCB 95	2.02	1.61	0.00	0.13	1.936	human	plasma	1.78	2.41
37680-73-2	PCB 101	2.04	1.61	0.00	0.13	1.936	human	plasma	1.70	2.42
38380-01-7	PCB 99	2.03	1.61	0.00	0.13	1.936	human	plasma	2.38	2.41
32598-14-4	PCB 105	2.04	1.59	0.00	0.11	1.936	human	plasma	2.11	2.43
31508-00-6	PCB 118	2.06	1.59	0.00	0.11	1.936	human	plasma	2.25	2.43
51908-16-8	PCB 146	2.19	1.74	0.00	0.11	2.059	human	plasma	2.48	2.47
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	human	plasma	2.43	2.47
39635-35-3	PCB 159	2.20	1.72	0.00	0.09	2.059	human	plasma	2.56	2.48
38380-08-4	PCB 156	2.21	1.72	0.00	0.09	2.059	human	plasma	2.49	2.48
52663-72-6	PCB 167	2.21	1.72	0.00	0.09	2.059	human	plasma	2.08	2.48
35065-28-2	PCB 138	2.18	1.74	0.00	0.11	2.059	human	plasma	2.04	2.47
35065-30-6	PCB 170	2.33	1.87	0.00	0.09	2.181	human	plasma	2.56	2.51
35065-29-3	PCB 180	2.29	1.87	0.00	0.09	2.181	human	plasma	2.43	2.51
52663-68-0	PCB 187	2.31	1.87	0.00	0.09	2.181	human	plasma	2.57	2.51
52663-69-1	PCB 183	2.30	1.87	0.00	0.09	2.181	human	plasma	2.20	2.51
52663-75-9	PCB 199	2.46	2.00	0.00	0.06	2.303	human	plasma	2.42	2.55
72-55-9	p,p'-DDE	1.80	1.40	0.06	0.14	2.053	human	plasma	2.23	2.46
<i>brain</i>										
107-19-7	propargyl alcohol	0.41	0.57	0.38	0.59	0.504	rat	blood	-0.23	0.00
107-13-1	acrylonitrile	0.30	0.83	0.03	0.30	0.502	rat	blood	-0.40	0.05
	1-(3-fluoropropyl)-2-nitroimidazole	0.76	1.64	0.00	0.76	1.171	rat	blood	-0.24	0.07
	1-(8-fluorooctyl)-2-nitroimidazole	0.75	1.62	0.00	0.76	1.875	rat	blood	-0.17	0.55
92817-10-2	16alpha-fluoroestradiol	1.85	2.39	0.96	1.06	2.237	rat	blood	-0.30	0.22
5400-60-2	4-fluoroantipyrine	1.38	1.60	0.00	1.25	1.522	rat	blood	-0.05	0.07
129-81-7	4-iodoantipyrine	2.10	2.05	0.00	1.31	1.743	rat	blood	-0.10	0.21
25526-93-6	alovudine	1.68	1.95	0.51	1.72	1.644	rat	blood	-0.59	-0.01
66893-81-0	alpha-(4-pyridyl-1-oxide)-N-tert-butyl nitrone	0.75	1.49	0.00	1.25	1.554	rat	blood	-0.38	0.03

60-80-0	antipyrine	1.30	1.83	0.00	1.37	1.485	rat	blood	-0.10	0.00
84379-13-5	bretazenil	2.39	2.58	0.00	1.82	2.701	rat	blood	-0.09	0.33
58-08-2	caffeine	1.50	1.82	0.08	1.25	1.363	rat	blood	-0.06	0.00
298-46-4	carbamazepine	2.15	2.18	0.76	0.88	1.811	rat	blood	0.00	0.32
154-93-8	carmustine	0.74	2.00	0.13	0.88	1.389	rat	blood	-0.52	0.02
486-56-6	cotinine	1.05	1.49	0.00	1.38	1.387	rat	blood	-0.32	-0.01
78755-81-4	flumazenil	1.74	2.33	0.01	1.46	2.088	rat	blood	-0.29	0.17
	fluoromisonidazole	1.02	1.88	0.32	1.29	1.230	rat	blood	-0.01	-0.01
604-75-1	oxazepam	2.35	1.10	0.45	1.60	1.992	rat	blood	0.61	0.34
103-90-2	paracetamol	1.06	1.63	1.04	0.86	1.172	rat	blood	-0.34	-0.01
611-59-6	paraxanthine	1.93	1.84	0.37	1.38	1.222	rat	blood	0.06	-0.01
60-80-0	antipyrine	1.30	1.83	0.00	1.37	1.485	rat	blood	-0.10	0.00
3376-24-7	phenyl-N-tert-butyltrione	0.73	1.03	0.01	0.49	1.536	rat	blood	0.05	0.73
2078-54-8	propofol	0.82	0.88	0.32	0.51	1.621	rat	blood	0.48	0.62
61413-54-5	rolipram	1.53	1.54	0.39	1.09	2.141	rat	blood	0.61	0.45
119054-09-0	tertbutylchlorambucil	1.19	1.49	0.03	0.89	2.826	rat	blood	1.00	1.16
5630-53-5	tibolone	1.77	2.60	0.22	1.14	2.579	rat	blood	0.40	0.42
7732-18-5	water	0.00	0.45	0.82	0.35	0.167	rat	blood	-0.04	-0.01
1088-11-5	nordazepam	2.21	1.60	0.26	1.13	1.933	rat	blood	0.50	0.51
90058-29-0	alaptide	1.17	1.54	0.48	1.20	1.391	rat	plasma	-0.26	-0.03
60-80-0	antipyrine	1.30	1.83	0.00	1.37	1.485	rat	plasma	-0.22	-0.06
94-25-7	butyl 4-aminobenzoate	1.01	1.20	0.29	0.75	1.595	rat	plasma	0.42	0.62
58-08-2	caffeine	1.50	1.82	0.08	1.25	1.363	rat	plasma	0.12	-0.05
486-56-6	cotinine	1.05	1.49	0.00	1.38	1.387	rat	plasma	-0.38	-0.07
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	0.03	0.96
938-73-8	ethenzamide	0.91	1.51	0.49	0.80	1.313	rat	plasma	-0.05	0.14
94-09-7	ethyl 4-aminobenzoate	1.03	1.31	0.31	0.69	1.313	rat	plasma	0.27	0.39
846-49-1	lorazepam	2.51	1.28	0.45	1.63	2.114	rat	plasma	0.47	0.77
69-65-8	mannitol	0.84	2.26	0.86	1.79	1.306	rat	plasma	-1.60	-0.08
2078-54-8	propofol	0.82	0.88	0.32	0.51	1.621	rat	plasma	0.91	0.82
94-12-2	propyl 4-aminobenzoate	1.02	1.20	0.31	0.73	1.454	rat	plasma	0.55	0.52
129-00-0	pyrene	2.81	1.71	0.00	0.28	1.585	rat	plasma	0.23	1.26
3056-17-5	stavudine	1.26	1.76	0.75	1.61	1.563	rat	plasma	-0.48	-0.08
83-67-0	theobromine	1.50	1.60	0.50	1.38	1.222	rat	plasma	-0.30	-0.08
30516-87-1	zidovudine	1.83	1.70	0.47	1.83	1.819	rat	plasma	-0.72	-0.06
58-15-1	aminopyrine	1.68	1.74	0.00	1.60	1.866	rat	serum	0.00	0.07
80-05-7	bisphenol A	1.61	1.56	0.99	0.91	1.864	rat	serum	-0.12	0.53
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	serum	0.52	0.96
1622-62-4	flunitrazepam	2.10	2.15	0.00	1.48	2.163	rat	serum	0.06	0.59
846-49-1	lorazepam	2.51	1.28	0.45	1.63	2.114	rat	serum	0.41	0.77
604-75-1	oxazepam	2.35	1.10	0.45	1.60	1.992	rat	serum	0.55	0.67
54-95-5	pentylene-tetrazole	0.90	1.24	0.00	1.02	1.050	rat	serum	-0.03	-0.06
28911-01-5	triazolam	2.64	2.20	0.00	1.65	2.327	rat	serum	0.57	0.75
<i>liver</i>										
	1-(3-fluoropropyl)-2-nitroimidazole	0.76	1.64	0.00	0.76	1.171	rat	blood	0.92	0.03
	1-(8-fluorooctyl)-2-nitroimidazole	0.75	1.62	0.00	0.76	1.875	rat	blood	0.47	0.40
2051-62-9	PCB 3	1.50	1.05	0.00	0.18	1.447	rat	blood	0.00	0.79
2050-68-2	PCB 15	1.65	1.18	0.00	0.16	1.569	rat	blood	0.48	0.84
33284-52-5	PCB 80	1.96	1.44	0.00	0.11	1.814	rat	blood	0.78	0.93
37680-73-2	PCB 101	2.04	1.61	0.00	0.13	1.936	rat	blood	0.78	0.93
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	rat	blood	1.08	0.97
13551-89-8	fluoromisonidazole	1.02	1.88	0.32	1.29	1.230	rat	blood	0.28	-0.05
123-31-9	hydroquinone	1.06	1.27	1.06	0.57	0.834	rat	blood	-0.12	0.03
80-05-7	bisphenol A	1.61	1.56	0.99	0.91	1.864	rat	blood	0.16	0.29
51333-22-3	budesonide	2.33	3.03	0.40	2.17	3.268	human	plasma	0.94	0.59

439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	0.65	0.82
1622-62-4	flunitrazepam	2.10	2.15	0.00	1.48	2.163	rat	plasma	0.57	0.54
846-49-1	lorazepam	2.51	1.28	0.45	1.63	2.114	rat	plasma	1.08	0.73
129-00-0	pyrene	2.81	1.71	0.00	0.28	1.585	rat	plasma	0.37	1.05
<i>lung</i>										
33671-46-4	clotiazepam	2.06	1.62	0.00	1.37	2.280	rat	blood	1.04	0.03
123-31-9	hydroquinone	1.06	1.27	1.06	0.57	0.834	rat	blood	-0.16	-0.10
146-22-5	nitrazepam	2.30	1.53	0.33	1.43	1.985	rat	blood	0.26	-0.16
51333-22-3	budesonide	2.33	3.03	0.40	2.17	3.268	rat	plasma	0.94	0.10
486-56-6	cotinine	1.05	1.49	0.00	1.38	1.387	rat	plasma	-0.18	-0.06
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	0.52	0.35
846-49-1	lorazepam	2.51	1.28	0.45	1.63	2.114	rat	plasma	0.44	0.23
938-73-8	ethenzamide	0.91	1.51	0.49	0.80	1.313	rat	plasma	-0.03	-0.12
129-00-0	pyrene	2.81	1.71	0.00	0.28	1.585	rat	plasma	0.35	0.57
<i>muscle</i>										
33284-52-5	PCB 80	1.96	1.44	0.00	0.11	1.814	rat	blood	0.00	0.24
2051-62-9	PCB 3	1.50	1.05	0.00	0.18	1.447	rat	blood	0.00	0.14
2050-68-2	PCB 15	1.65	1.18	0.00	0.16	1.569	rat	blood	0.30	0.17
37680-73-2	PCB 101	2.04	1.61	0.00	0.13	1.936	rat	blood	0.00	0.25
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	rat	blood	0.60	0.28
80-07-9	p,p'-dichlorodiphenyl sulfone	1.88	2.20	0.00	0.54	1.850	rat	blood	0.58	0.00
15569-85-4	cotinine	1.05	1.49	0.00	1.38	1.387	rat	plasma	-0.05	-0.08
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	0.15	0.38
938-73-8	ethenzamide	0.91	1.51	0.49	0.80	1.313	rat	plasma	-0.09	-0.05
129-00-0	pyrene	2.81	1.71	0.00	0.28	1.585	rat	plasma	0.20	0.36
<i>skin</i>										
35065-27-1	PCB 153	2.18	1.74	0.00	0.11	2.059	rat	blood	1.48	1.41
37680-73-2	PCB 101	2.04	1.61	0.00	0.13	1.936	rat	blood	0.85	1.35
33284-52-5	PCB 80	1.96	1.44	0.00	0.11	1.814	rat	blood	0.85	1.33
2050-68-2	PCB 15	1.65	1.18	0.00	0.16	1.569	rat	blood	1.00	1.19
2051-62-9	PCB 3	1.50	1.05	0.00	0.18	1.447	rat	blood	1.00	1.12
124-18-5	decane	0.00	0.00	0.00	0.00	1.518	rat	blood	0.68	1.49
80-07-9	p,p'-dichlorodiphenyl sulfone	1.88	2.20	0.00	0.54	1.850	rat	blood	1.21	0.73
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rat	plasma	0.54	1.02
938-73-8	ethenzamide	0.91	1.51	0.49	0.80	1.313	rat	plasma	0.02	0.11
33671-46-4	clotiazepam	2.06	1.62	0.00	1.37	2.280	rabbit	plasma	0.15	1.10
439-14-5	diazepam	2.08	1.55	0.00	1.28	2.074	rabbit	plasma	0.20	1.02
146-22-5	nitrazepam	2.30	1.53	0.33	1.43	1.985	rabbit	plasma	0.20	0.63

Table S7. PP-LFER descriptors of 15 probe chemicals used for analysis of relative sorptive capacity of different matrices.

	E	S	A	B	V
trichloroethylene	0.52	0.37	0.08	0.03	0.715
hexachlorobutadiene	1.02	0.42	0.00	0.16	1.321
lindane	1.45	1.28	0.00	0.50	1.580
hexachlorobenzene	1.49	0.99	0.00	0.00	1.451
benzo[<i>a</i>]pyrene	3.63	1.96	0.00	0.37	1.954
PCB 28	1.76	1.33	0.00	0.15	1.691
PCB 180	2.29	1.87	0.00	0.09	2.181
atrazine	1.51	1.23	0.00	1.05	1.620
metolachlor	1.15	1.01	0.07	1.38	2.281
diuron	1.50	1.59	0.50	0.73	1.599
2,4,5-trichloroaniline	1.24	1.15	0.30	0.14	1.183
<i>n</i> -butylparaben	0.86	1.35	0.69	0.45	1.555
4- <i>n</i> -nonylphenol	0.77	0.88	0.55	0.37	2.043
bisphenol A	1.61	1.56	0.99	0.91	1.864
caffeine	1.50	1.82	0.08	1.25	1.363

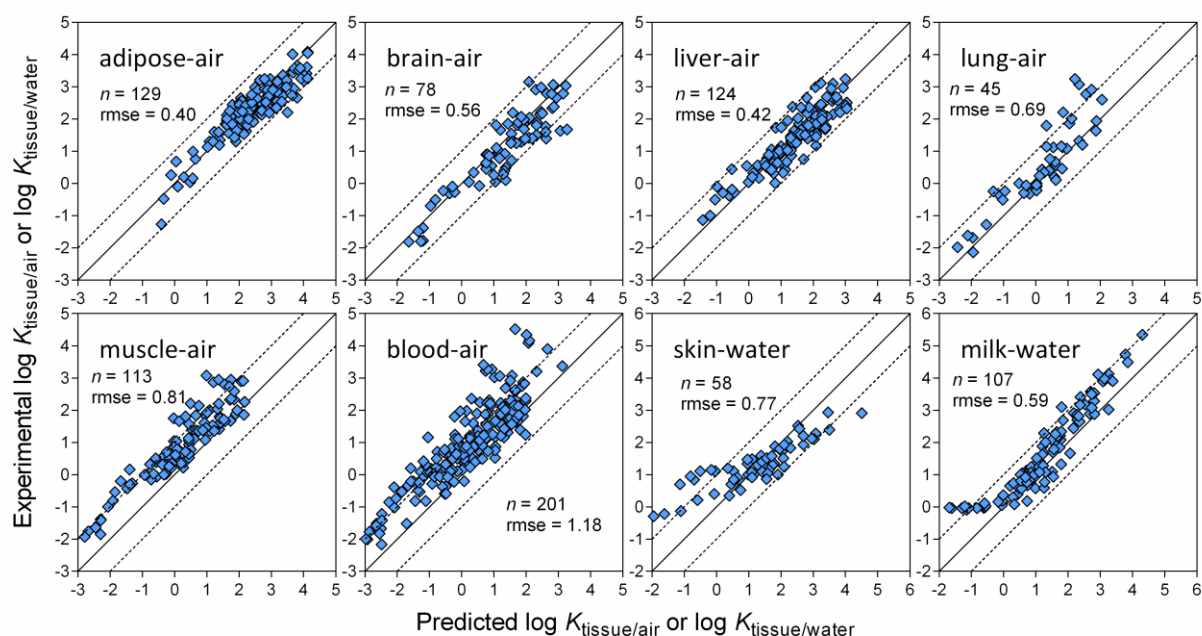


Figure S1. Comparison of in-vitro partition coefficients with values predicted from the lipid-octanol model. The solid line indicates the 1:1 agreement and the dashed lines a difference of 1 log unit.

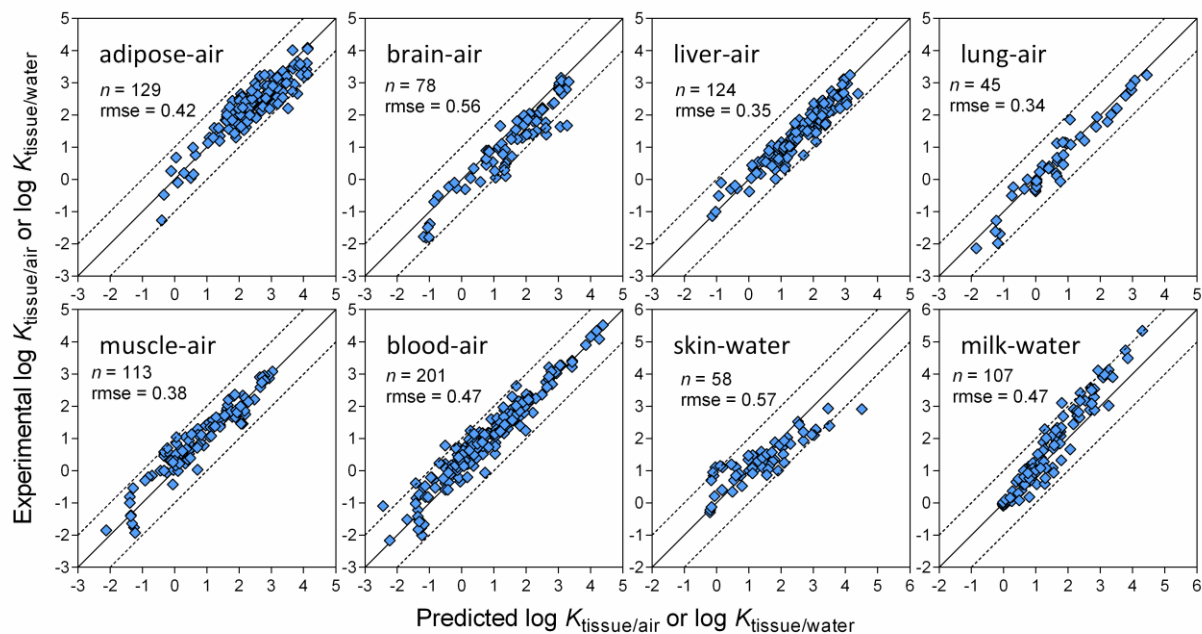


Figure S2. Comparison of in-vitro partition coefficients with values predicted from the “lipid-octanol + water” model. The solid line indicates the 1:1 agreement and the dashed lines a difference of 1 log unit.

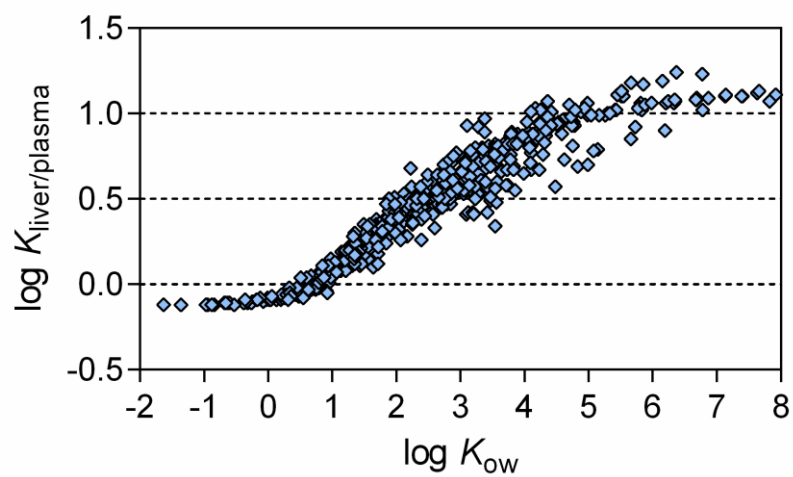


Figure S3. Log of liver-plasma partition coefficients against $\log K_{\text{ow}}$. Liver-plasma partition coefficients are calculated with the PP-LFER-composition-based approach. K_{ow} values were calculated with the PP-LFER model. PP-LFER descriptors for > 600 diverse compounds were used to cover a large chemical domain.

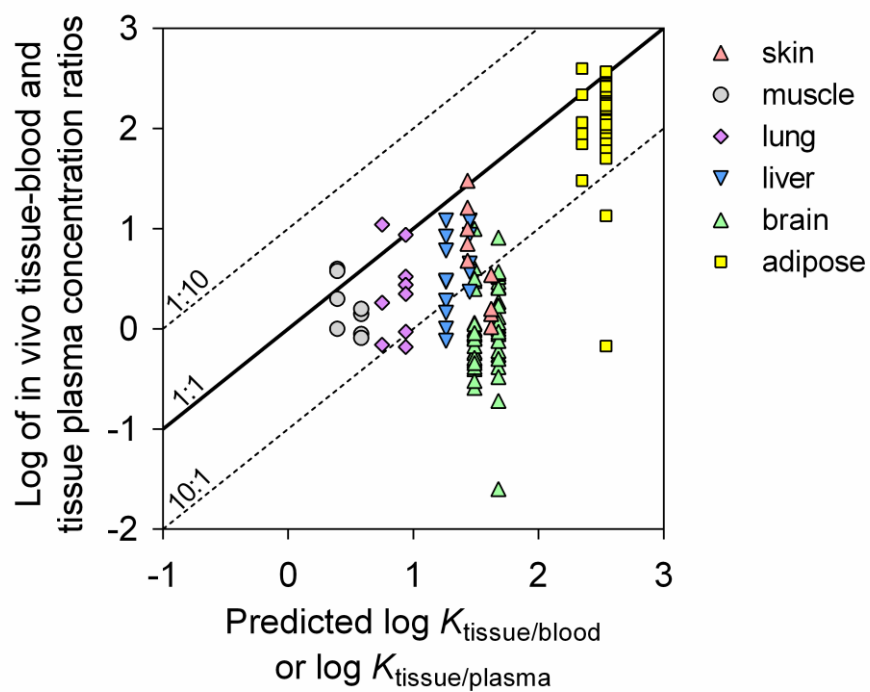


Figure S4. Predictions of in vivo tissue-blood and tissue-plasma distribution ratios using the lipid-octanol model.

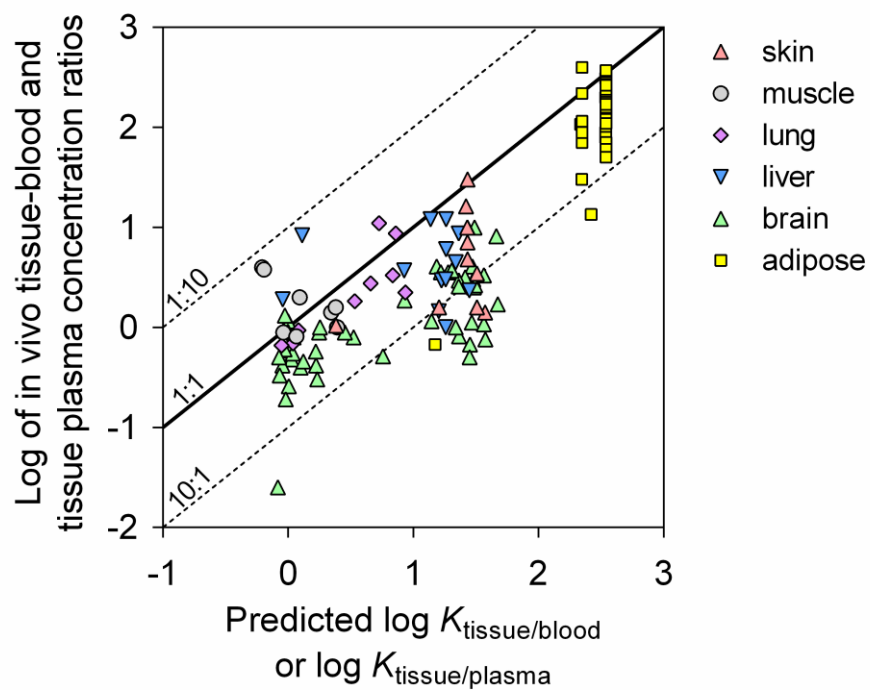


Figure S5. Predictions of in vivo tissue-blood and tissue-plasma distribution ratios using the "lipid-octanol + water" model.

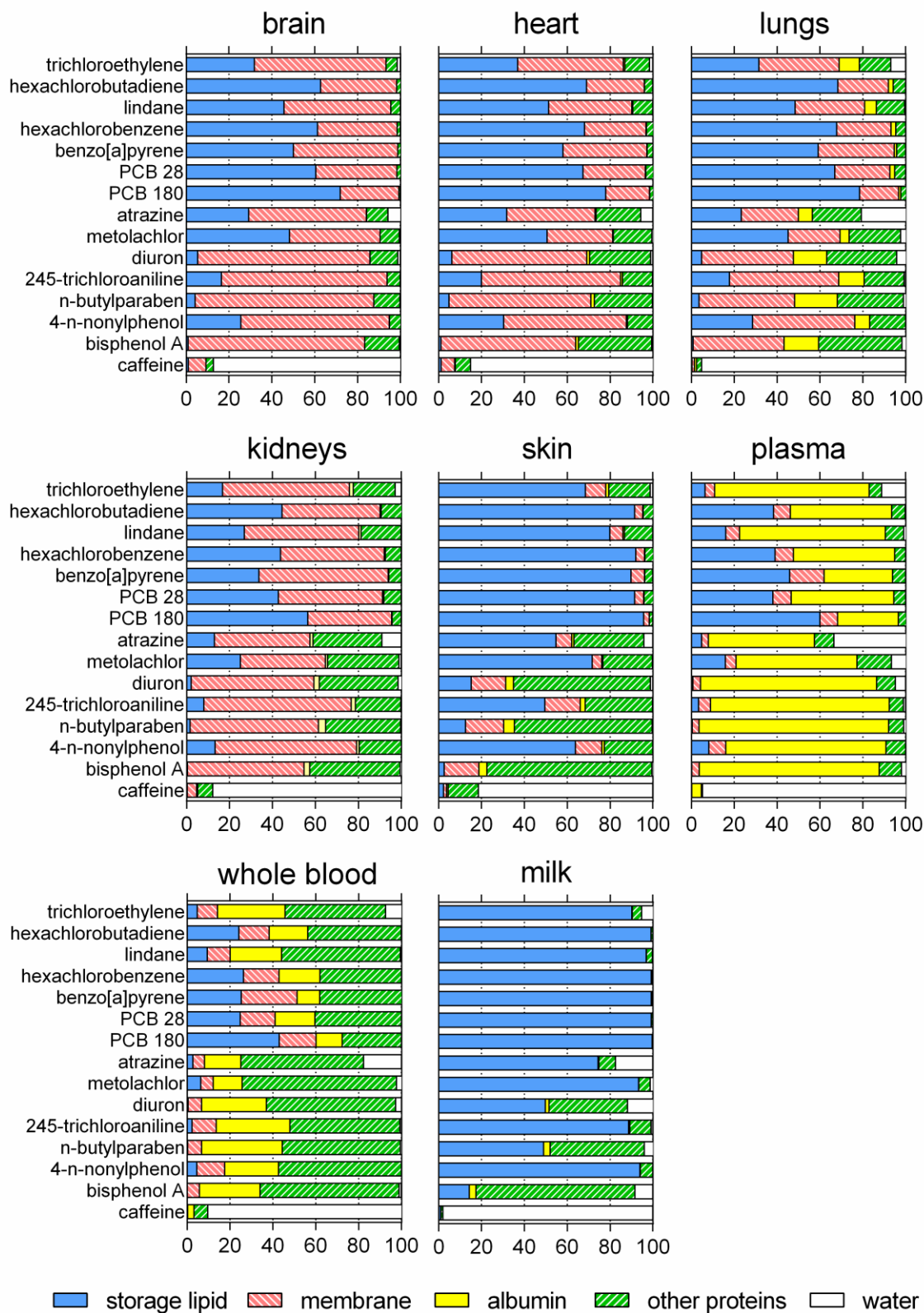


Figure S6. Predicted % contributions of tissue components to the overall sorptive capacity for selected chemicals.

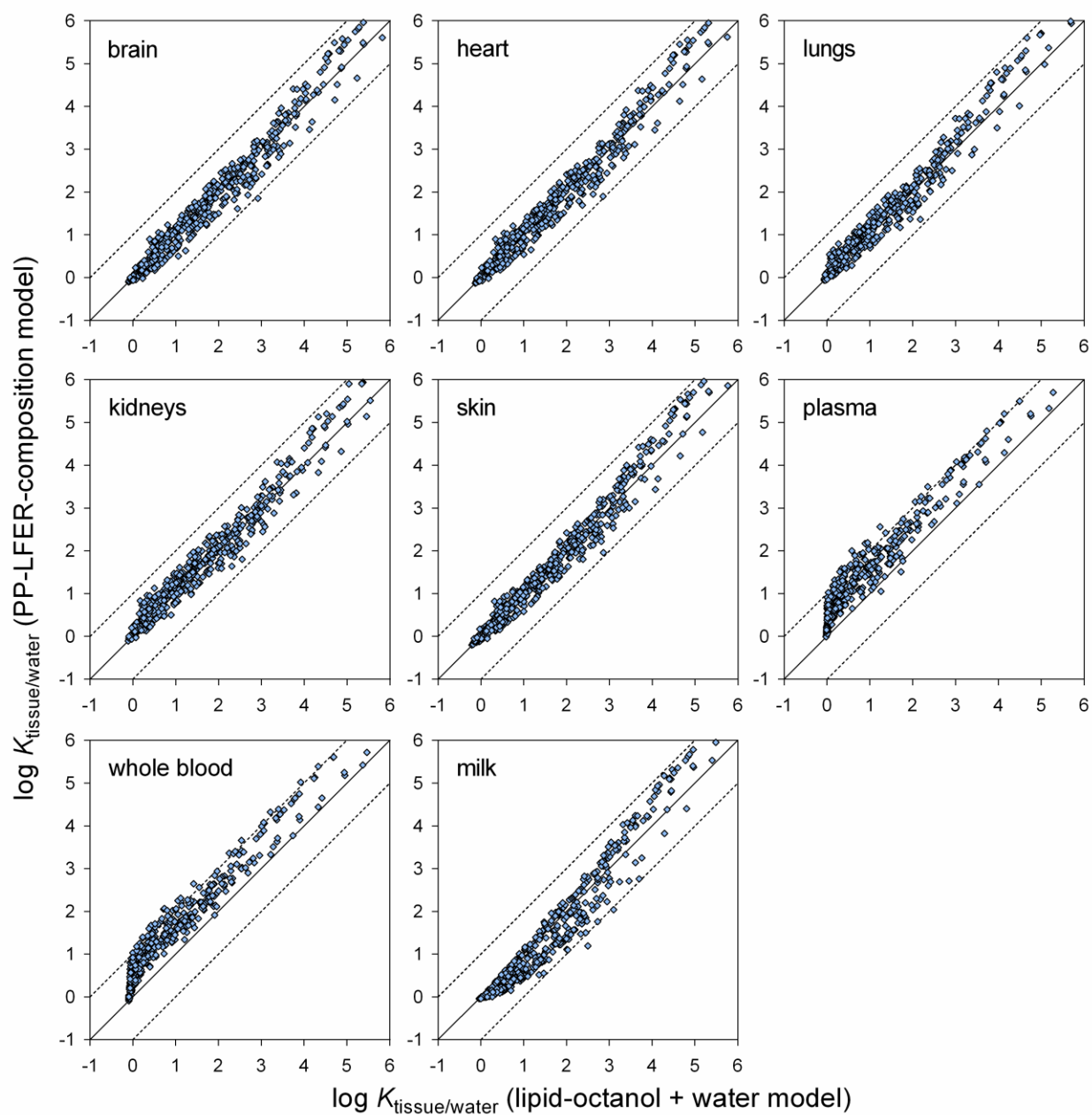


Figure S7. Comparison of $K_{\text{tissue/water}}$ predicted with PP-LFER-composition and “lipid-octanol + water” models.

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